The Generalized MCM Polynomial Is a Permutation Polynomial

1 Statement

Let $F = \mathbb{F}_{2^n}$, and let $0 < k < n$. For $b \in \mathbb{F}_2$, define

$$\psi_b(x) = \sum_{i=0}^{k-1} x^{2^i} + b\,\mathrm{Tr}(x)$$

and define the generalized Muller–Cohen–Matthews polynomial by

$$\mathrm{MCM}_b(X) = \left(\sum_{i=0}^{k-1} X^{2^i} + b\,\mathrm{Tr}(X)\right)^{2^k+1} X^{(2^n-1)-2^k}.$$

Here $\mathrm{Tr}(x) = \sum_{i=0}^{n-1} x^{2^i}$ is the absolute trace of F over $\mathbb{F}_2$. This is the polynomial called `genMCMpol` (with arguments n, k, and b) in the Lean development.

Theorem 1. *Suppose that* $\gcd(k, n) = 1$ *and that* k *is odd. Then* MCM_0 *induces a permutation of* F*.*

The proof is the $b = 0$ specialization of the transfer theorem formalized as `dobbertin_theorem4`. We state precisely the part of that theorem used below and then verify its arithmetic hypotheses.

2 The transfer theorem

For positive integers k, n with $\gcd(k, n) = 1$, choose integers k^{-1} and n' satisfying

$$0 < k^{-1} < n, \qquad k^{-1}k = n'n + 1. \tag{1}$$

For $a \in \mathbb{F}_2$, put

$$\phi_a(x) = \sum_{i=0}^{k^{-1}-1} x^{2^{ik}} + a\,\mathrm{Tr}(x).$$

Lemma 2 (Kasami parity criterion)**.** *Assume* (1) *and*

$$k^{-1} + an \equiv 1 \pmod 2. \tag{2}$$

Then the generalized Kasami polynomial Kas_a *is a permutation polynomial of* F*.*

Proof idea. The formal lemma `parity_implies_permutation` proves that the evaluation map of Kas_a is injective and then uses finiteness of F to obtain surjectivity. Its injectivity argument translates equality of two values of Kas_a into two instances of the associated Kasami equation

$$cz^{2^k+1} = R_a(z),$$

where R_a is the equation's right-hand side. The zero and nonzero input cases are treated separately so that the required nonzero admissibility condition is retained when $c \neq 0$.

The key uniqueness lemma splits into three cases. If $c = 0$, a separate zero-right-hand-side result gives the unique solution. If $c \neq 0$ and $c = \gamma^{2^k+1} + \gamma$ for some γ, the odd parity (2) excludes a solution in the zero fiber of the auxiliary gamma polynomial. The remaining fiber is controlled by the Delta and Ell identities; two distinct solutions would force both a parity identity $e_0 + e_1 = 1$ and the vanishing of the corresponding e-terms, a contradiction. Finally, when no such γ exists, uniqueness follows from the linearized equation. Thus every admissible Kasami-equation fiber has at most one solution, proving injectivity of Kas_a. □

Theorem 3 (Dobbertin transfer theorem)**.** *Assume* (1)*, and suppose that* $a, b \in \mathbb{F}_2$ *satisfy*

$$k^{-1} + an \equiv 1 \pmod 2, \tag{3}$$

$$b \equiv n' + ak \pmod 2. \tag{4}$$

Then ψ_b *and* ϕ_a *are inverse permutations of* F*. Moreover, if* Kas_a *denotes the associated generalized Kasami polynomial, then*

$$\mathrm{Kas}_a \circ \psi_b = \iota \circ \mathrm{MCM}_b,$$
$$\mathrm{MCM}_b \circ \phi_a = \iota \circ \mathrm{Kas}_a,$$

where $\iota(x) = x^{-1}$*, with* $\iota(0) = 0$*. If the parity condition* (3) *holds, then Lemma 2 shows that* Kas_a *is a permutation polynomial; consequently,* MCM_b *is a permutation polynomial.*

For context, the formal proof of this transfer theorem has two main inputs. First, Frobenius periodicity $x^{2^n} = x$ and trace invariance show that ψ_b and ϕ_a are mutual inverses under (3)–(4). Second, the parity condition makes the generalized Kasami polynomial a permutation by the fiber-uniqueness argument in Lemma 2. The second displayed functional identity then makes $\mathrm{MCM}_b \circ \phi_a$ a composition of permutations. Since ϕ_a is itself a permutation, MCM_b is a permutation as well.

3 Proof of Theorem 1

Because $\gcd(k, n) = 1$, k has a multiplicative inverse modulo n. Choose k^{-1} with

$$0 < k^{-1} < n, \qquad k^{-1}k \equiv 1 \pmod n.$$

Set

$$n' = \frac{k^{-1}k}{n}.$$

The modular-inverse congruence says that $k^{-1}k$ has remainder 1 upon division by n, so the division algorithm gives

$$k^{-1}k = n'n + 1. \tag{5}$$

Thus the first set of hypotheses of Theorem 3 is satisfied.

Choose $a \in \mathbb{F}_2$ to be the parity of n'; equivalently,

$$a \equiv n' \pmod 2. \tag{6}$$

Since k is odd,

$$k \equiv 1 \pmod 2. \tag{7}$$

We verify the two parity relations required by the transfer theorem.

Multiplying (7) by k^{-1} and using (5) yields

$$k^{-1} \equiv k^{-1}k \equiv n'n + 1 \pmod 2.$$

Together with (6), this gives

$$k^{-1} + an \equiv (n'n + 1) + n'n = 2n'n + 1 \equiv 1 \pmod 2.$$

Hence (3) holds.

For the second relation, (6) and (7) imply

$$ak \equiv n'k \equiv n' \pmod 2.$$

Therefore

$$n' + ak \equiv n' + n' \equiv 0 \pmod 2.$$

This is exactly (4) with $b = 0$.

All hypotheses of Theorem 3 now hold for the data $(n, k, k^{-1}, n', a, b = 0)$. Its conclusion states that MCM_0 is a permutation polynomial of F, which proves Theorem 1. $\square$

4 Formalization Correspondence

The construction of k^{-1} is `Nat.exists_mul_mod_eq_one_of_coprime`. The quotient n' is `nPrime`, equation (5) is `hprod`, and the element a is the Lean value `a : Fin 2`. The two parity calculations are `hqpar` and `hbeta`. Lemma 2 corresponds to `parity_implies_permutation`: it uses `kasamiEquation_unique` to prove injectivity and `Finite.injective_iff_surjective` to conclude bijectivity. Finally, the formal theorem invokes `dobbertin_theorem4` with $b = 0$ and selects its final conclusion, `IsPermutationPolynomial (genMCMpol n k 0)`.